

AI PSYCHIATRIST ENGINE

COGNITIVE FLOW & FOCUS TRACKING | README & QUICK-START GUIDE

LOCK. TRACK. ANALYZE.

A CPU-optimized computer-vision pipeline for analyzing player movement cues such as hesitation, spatial scanning, and estimated cognitive fatigue (La Pausa) from match footage.

Repository overview

This repository contains the source code for Phase 3 of the Remontada Ecosystem. Its intended workflow combines target selection, ball detection, player telemetry, cognitive-state heuristics, and HUD-style video rendering. Replace LINK_TO_YOUR_MASTER_REPO_HERE with the URL of the master repository.

The Master Repository / whitepaper should provide the full derivations for Circular Standard Deviation (heading variance), Bayesian Fault Isolation, exponential cognitive decay (e^{-kt}), and the 10-frame possession buffer.

Interpretation caveat: these outputs are computer-vision-derived proxies, not direct measurements or diagnoses of a player's psychological state. Validate them against labeled footage and treat them as scouting aids, not clinical conclusions.

Execution & rendering flow

1. Target Lock Draw player ROI	2. Ball Tracker Class 32 detection	3. Telemetry BotSORT + heading filter
6. H.264 Output Rendered video	5. HUD Rendering Vision cones + bars	4. Cognitive Engine La Pausa + decay estimates

Pipeline summary: target selection → ball extraction → player telemetry → heuristic cognitive indicators → HUD overlays → H.264 video export.

1. Repository Structure

```
3-AI-Psychiatrist/  
■■■ main_2.py           # Four-stage pipeline: lock, track, analyze, render  
■■■ ball_tracker.py     # Class 32 ball extraction + frame memory buffer  
■■■ main_engine.py      # Orchestrates vision, tactics, and cognitive logic  
■■■ cognitive_decay.py  # Decay math, La Pausa validation, vector matching  
■■■ tactical_blueprint.py # Adjusted-value formulas and lane interference  
■■■ verdict_logic.py    # Bayesian fault isolation and tactical reset  
■■■ README.md           # Quick-start documentation
```

The project description refers to a custom BotSORT configuration file (botsort_custom.yaml). If it is generated automatically, confirm where it is written; if it is required at runtime, make sure the setup step creates it before tracking begins.

2. Quick Start & CLI Usage

Before running

Use a Python environment with the project’s required computer-vision and tracking packages installed, and ensure the input match video is accessible. The exact dependency list was not included in the supplied README; add a requirements.txt or environment.yml so setup is reproducible.

1. Configure the workspace. Open main_2.py and set WORKSPACE_DIR and INPUT_VIDEO to the correct local paths.
2. Run the pipeline. From a terminal in the project directory, execute the command below.

```
python main_2.py
```

3. Stage 1 — Target Acquisition

The video opens on a freeze-frame selection window. Use the keyboard controls to locate the desired player, draw a region of interest (ROI), and save the selected target for downstream tracking.

Key	Action
D / A	Skip forward/back by 30 frames (confirm direction in the implementation)
W / S	Skip by 5 frames
. / ,	Move by one micro-frame / single-frame step
SPACE	Draw a bounding box around the target player
F	Save targets to saved_targets3.json and continue

After target acquisition, the described pipeline tracks the ball, builds a cognitive telemetry timeline, and renders the HUD output video. Exact output paths and codec availability depend on the implementation and local environment.

4. Cinematic HUD Legend

Overlay	Meaning in the supplied design	Interpretation caution
Focus bars (0-100%)	Heuristic score decreases during high-speed movement (>6.0 m/s) or duels and may recover during scanning/jogging.	Not a direct measure of attention or fatigue; calibrate against data.
Gold vision cone	Shown when circular standard deviation of the player's trajectory/heading exceeds 10.0° as a spatial-scanning cue.	Heading variance alone does not prove a shoulder check or awareness.
Possession / La Pausa timer	Tracks estimated possession time; "La Pausa" appears when latency >1.5 s and space value rises.	Requires reliable possession, timing, and space-value estimates.
Hesitation indicator	Red text appears when the focus heuristic falls below 65%.	A threshold alert is a model output, not a psychological diagnosis.

5. Technical Concepts Referenced

Circular Standard Deviation: a directional-statistics measure for angular spread, appropriate for headings that wrap around 0°/360°.

Bayesian Fault Isolation: a probabilistic framework for ranking likely causes of an observed tracking or tactical fault; its validity depends on the model and priors.

Exponential decay: a time-dependent heuristic such as $C(t) = C_0 e^{-kt}$; parameter k and the meaning of $C(t)$ must be empirically defined.

Possession buffer: a short rolling frame window intended to stabilize possession-related estimates across noisy detections.

These are documentation-level descriptions. The actual implementation should be checked before treating any formula, threshold, or runtime behavior as verified.

6. Troubleshooting

- Video fails to open: Check the input path, file permissions, and codec support.
- Model/tracker errors: Verify installed packages, model weights, tracker configuration, and the expected YOLO class mapping for the ball.
- Target IDs are not retained: Check that `saved_targets3.json` is created in the expected directory and is readable on the next run.
- Output video is missing: Confirm the output path, available disk space, and H.264 encoder support.
- Performance is slow: Confirm model size, input resolution, frame sampling, and whether CPU-only inference is expected.

7. License

Released under the MIT License, according to the supplied project description. Include a LICENSE file containing the intended license text before distribution.

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A computer-vision workflow for studying observable match behavior through telemetry and video overlays—not a clinical or psychological diagnostic system.